

# Role of lentils (*Lens culinaris* L.) in human health and nutrition: a review

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**Abstract** Humans have known lentils (*Lens culinaris* L.) since the dawn of civilization. The current work is a comprehensive review of lentils composition, nutritional value, and health benefits. The article addresses major proteins identified in lentils and their bioactive peptides, including lectins, defensins, and protease inhibitors. In addition, this review discusses the complex carbohydrate fractions in lentils, particularly the resistant starches, oligosaccharides, and dietary fibers with emphasis on their biomedical properties. Also, the current review discusses minerals and vitamins as well as the non-nutritive bioactive phytochemicals of lentils which add to the promising potential for clinical applications in the management and prophylaxis of several chronic human illnesses. The article finds out that various potential health benefits have been described for lentils such as anticarcinogenic, blood pressure-lowering, hypocholesterolemic and glycemic load-lowering effects. The proposed mechanisms behind each health benefit are discussed.

**Keywords** Lentils (*Lens culinaris* L.) · Functional foods · Nutritional value · Health · Polyphenols · Bioactive peptides

## Introduction

Nowadays, pulses are gaining more interest in the field of developing healthy and functional foods. Numerous articles [1–6] critically reviewed the macro and micro nutritional components and bioactive phytochemicals of pulses and their beneficial health uses. Among these pulses, lentils had gained much interest with respect to their unique nutritional and functional characteristics. An accumulating body of evidence is growing nowadays on the health-improving and disease-preventing impact of lentils. Thus, the current article is a comprehensive review that appraises the presently available information on their composition, nutritive value, and functionality.

The lentil plant, *Lens culinaris* L., is a member of the Leguminosae family and constitutes one of the most important traditional dietary components [7]. FAO reported that world production of lentils was about 2.83 million metric tons for 2008, primarily coming from Canada (36.9 %) and India (28.7 %), followed by Nepal, China and Turkey [8].

Lentils have been incorporated into different world cuisines throughout the globe, especially the Mediterranean and Indian regions. Lentils have been classified among soft seed-coated pulses that require shorter cooking time, and thus have smaller losses in nutrients as compared to those with hard seed coat [9]. The shorter cooking time of lentils (23–26 min) in comparison with most other pulses (>70 min) makes lentils very convenient for human consumption [10, 11].

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## Composition of lentils

The nutritional composition and individual nutrients of whole and split lentils had been readily identified, and are summarized in Table 1 [12].

### Macronutrients

Lentils are considered to be a good source of proteins. This high protein content in lentils and other pulses makes them a significant food source for developing countries and low-income people [13].

Most of the pulse proteins are storage proteins, which are usually consumed by the germ during seed germination. In addition to providing essential and non-essential amino acids and carbon skeletons for the metabolic needs of the human body, lentils are sources for some storage proteins that are described as biologically active proteins. These proteins have been historically referred to as “anti-nutritional” compounds such as lectins and protease inhibitors (PIs) [14]. Further, the protein fraction of total lentil nitrogen or the protein nitrogen (PN) is considerably high (4.49 g N/100 g DM), representing about 89 % of total nitrogen, while the non-protein nitrogen (NPN) part accounts for the remaining part, a matter that indicates better nitrogen usability in lentils than in high NPN foods [14]. These characteristics of lentil proteins make lentils a source of proteinous edible films that could be used in food industries [15].

Total carbohydrates represent the major component of lentil seeds [16] with starches occupying most of the carbohydrate mass (Table 1). Among 23 pulse grains, starch yield percent from lentils is the second highest, up to 47.1 % [13]. Furthermore, lentils are a valuable source of total dietary fibers, with insoluble dietary fiber of approximately 93–99.7 % [17].

The total  $\alpha$ -galactosides or raffinose family oligosaccharides account for 53.0 % of the total sugars and oligosaccharides content in lentils [18]. In these oligosaccharides, stachyose represents the major oligosaccharide, followed by ciceritol and raffinose [14, 18]. The functional significance of these carbohydrates arises from their ability to work as selective promoters for the growth of beneficial gut microbes. Han found that among different pulses, ciceritol was uniquely identified in lentils and chickpeas, whereas verbascose was uniquely identified in lentils and peas [19].

Lentils have relatively low fat and, therefore, low energy content (Table 1) [12, 16]. Ryan and colleagues [20] found that lentil seeds contained a total fat of about 1.4 g/100 g, distributed unevenly over the fatty fractions as follows: saturated fatty acids (SFA), 16.7 %; monounsaturated fatty acids (MUFA), 23.7 % and polyunsaturated fatty acids (PUFA), 58.8 %.

### Micronutrients

Ash content of lentils is relatively high, with a range value of 3–5 g/100 g DM. Indeed, the mineral content of lentils is composed of relatively high levels of Mg, P, Ca and S. In addition, lentils have a low Na and relatively high K contents, with a K:Na ratio of about 30:1–90:1 [16, 21].

Iron (Fe) is also present in significant quantity in lentils [16, 21]. Because bioavailability of iron from lentils could be adversely affected by natural chelating agents present in pulses, this adverse effect could be minimized by cooking, germination and fermentation of lentils prior to ingestion [22]. In addition, lentils contain Zn, which ranges between 3.2 and 6.3 mg/100 g. In addition, other trace minerals including Cu, Mn, Mo, and B [21, 22].

Selenium content of lentils is varied according to soil content and agricultural practises. For example, lentils grown in Saskatchewan province/Canada contain 425–673  $\mu$ g Se/kg [23], which provides 80–120 % of the Se recommended dietary allowance (RDA) in just a 100 g of dry lentils. Thus, Se has drawn significant interest as a target crop for Se biofortification to be used as a strategic food-based solution for populations with Se deficiencies [24].

Lentils are a significant dietary source of a plethora of vitamins including folate, thiamin ( $B_1$ ) and riboflavin ( $B_2$ ) [12]. Other water-soluble vitamins have also been reported in lentils as follows: niacin; pantothenic acid and pyridoxine. In addition, vitamin E ( $\alpha$ ,  $\beta$  and  $\gamma$  tocopherols) was measured in lentils by Ryan et al. [20]. The  $\alpha$ -tocopherol and  $\beta$  and  $\gamma$ -tocopherols contents were 1.6 and 4.5 mg/100 g, respectively. This, however, is far from the RDA value of  $\alpha$ -tocopherol (15 mg/day). Further, phyloquinone (vitamin K) as reported by USDA [12] averages about 5  $\mu$ g/100 g, while the recommended daily value of vitamin K is about 80  $\mu$ g (Table 1). This low vitamin K content renders lentils safe in cardiovascular patients taking the anticoagulant warfarin.

Accordingly, lentils have been developed into nutritious acceptable snack bar in order to be included in the regular daily meal with improved acceptability. The developed product resulted in more dietary fiber, protein and iron, in addition to an approximate sevenfold increase in folate over three control commercial snack bars used in the study [25].

### Bioactive components

It is not surprising that bioactive phytochemicals have been identified and quantified in pulse seeds, making them major

**Table 1** Nutrient content of whole and split lentils, g/100 g FM

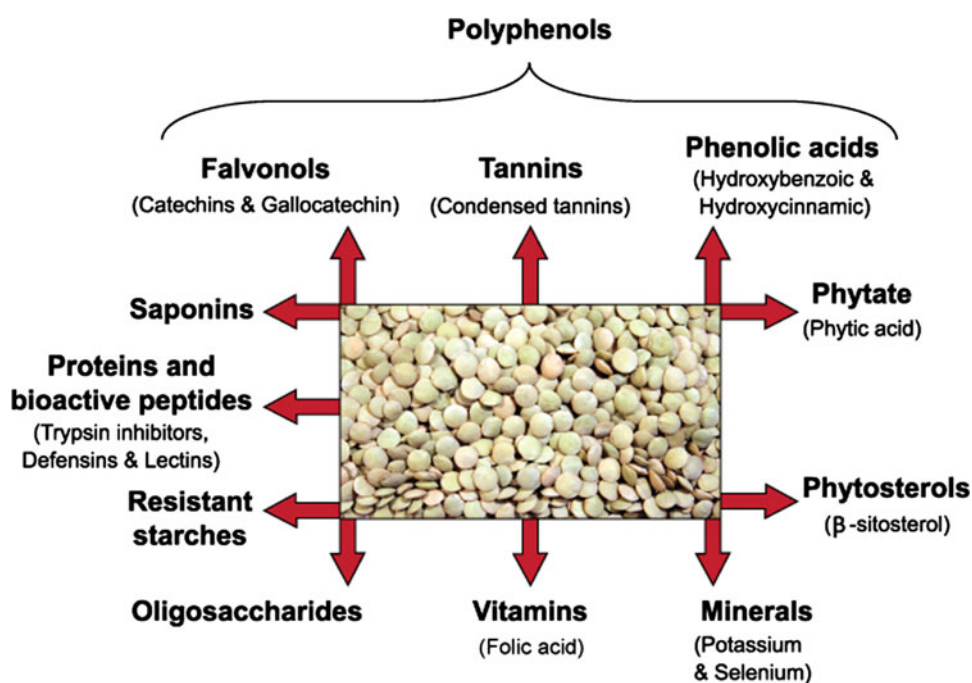
| Nutrient                       | Unit    | Whole lentils | Split lentils | Dietary reference value |
|--------------------------------|---------|---------------|---------------|-------------------------|
| Proximate                      |         |               |               |                         |
| Water                          |         | 10.4          | 11.8          | –                       |
| Energy                         | kJ      | 1,477         | 1,443.5       | 2,000                   |
| Protein                        | g       | 25.8          | 25            | 50                      |
| Total lipids (Fat)             | g       | 1.1           | 2.2           | 65                      |
| Ash                            | g       | 2.7           | 1.9           | –                       |
| Carbohydrates, by difference   | g       | 60.1          | 59.2          | 300                     |
| Fibers, total dietary          | g       | 30.5          | 10.8          | 25                      |
| Sugars, total                  | g       | 2.0           | –             | –                       |
| Sucrose                        | g       | 1.5           | –             | –                       |
| Glucose (Dextrose)             | g       | 0.0           | –             | –                       |
| Fructose                       | g       | 0.3           | –             | –                       |
| Lactose                        | g       | 0.0           | –             | –                       |
| Maltose                        | g       | 0.3           | –             | –                       |
| Galactose                      | g       | 0.0           | –             | –                       |
| Minerals                       |         |               |               |                         |
| Calcium, Ca                    | mg      | 56            | 41            | 1,000                   |
| Iron, Fe                       | mg      | 7.5           | 7.6           | 18                      |
| Magnesium, Mg                  | mg      | 122           | 72            | 400                     |
| Phosphorus, P                  | mg      | 451           | 294           | 1,000                   |
| Potassium, K                   | mg      | 955           | 578           | 3,500                   |
| Sodium, Na                     | mg      | 6             | 7             | 2,400                   |
| Zinc, Zn                       | mg      | 4.8           | 3.9           | 15                      |
| Copper, Cu                     | mg      | 0.5           | 1.3           | 2                       |
| Manganese, Mn                  | mg      | 1.3           | 1.4           | 2                       |
| Selenium, Se                   | µg      | 8.3           | 8.2           | 70                      |
| Vitamins                       |         |               |               |                         |
| Vitamin C, total ascorbic acid | mg      | 4.4           | 1.7           | 60                      |
| Thiamin                        | mg      | 0.9           | 0.5           | 1.5                     |
| Riboflavin                     | mg      | 0.2           | 0.1           | 1.7                     |
| Niacin                         | mg      | 2.6           | 1.5           | 20                      |
| Pantothenic acid               | mg      | 2.1           | 0.3           | 10                      |
| Pyridoxine (B6)                | mg      | 0.5           | 0.4           | 2                       |
| Folate, total DFE              | µg, DFE | 479           | 204           | 400                     |
| Vitamin A, IU                  | IU      | 39            | 58            | 5,000                   |
| Vitamin A, RAE                 | µg -RAE | 2             | 3             | –                       |
| Carotene, beta                 | µg      | 23            | 35            | –                       |
| Vitamin E (α-tocopherol)       | mg      | 0.5           | –             | 30 IU                   |
| Tocopherol (γ-tocopherol)      | mg      | 4.2           | –             | –                       |
| Vitamin K (phylloquinone)      | µg      | 5.0           | –             | 80                      |
| Choline, total                 | mg      | 96.4          | –             | –                       |
| Lipids                         |         |               |               |                         |
| Fatty acids, total SFA         | g       | 0.2           | 0.4           | 20                      |
| Fatty acids, total MUFA        | g       | 0.2           | 0.5           | –                       |
| Fatty acids, total PUFA        | g       | 0.5           | 1.1           | –                       |
| Cholesterol                    | mg      | 0.0           | 0.0           | 300                     |

Obtained from the United States Department of Agriculture (USDA) National Nutrient Database for Standard Reference, Release 23 (2010). (Accessed Dec. 2010) [12]

sources for phytochemicals in the diet [26]. Bioactive components and essential nutrients in lentil seeds could be categorized into different functional compounds according

to their chemical structure (Fig. 1), and the quantities of the main different bioactive phytochemicals of lentils are listed in Table 2.

**Fig. 1** Bioactive phytochemicals and essential nutrients of lentil seeds



## Polyphenols

Lentils have the highest total phenolic content (TPC) in comparison with six other common pulses [19, 27]. Among polyphenols, tannins and tannin-related compounds are principal in lentils, and mainly concentrated in the testa. Lentils are being among the richest leguminous seeds in their condensed tannin content, up to 915 mg/100 g [28]. TPC and antioxidant activity in lentils exhibit higher or compatible value with those of fruits or vegetables. In addition, TPC of lentils exhibited significant correlation with total antioxidant activity, implying that phenolic compounds are major antioxidant compounds in lentils [19, 29]. Recently, lentils were reported to score the highest values among 14 different types and varieties of pulses for simple polyphenols and the TPCs, with TPC of about 26 mg gallic acid equivalent (GAE)/100 g FM [30]. Recent reports indicated that catechin glucosides, procyanidin dimers, quercetin diglycoside, and *trans-p*-coumaric acid were the dominant phenolics in green lentils, while quercetin diglycoside, catechin, digallate procyanidin, and *p*-hydroxybenzoic were the dominant phenolics in red lentils [31, 32].

The distribution of phenolic compounds differs in the cotyledon and the lentil seed coat, with non-flavonoid phenolic compounds, such as free and combined hydroxy benzoic and hydroxycinnamic acids, located mainly in the cotyledon of lentils [32]. On the other hand, flavonoids, such as glycosides of flavonols and flavones, are mainly present in the seed coat of lentils. The coat also contains *trans*-resveratrol-3-*O*-glucoside, and large amounts of proanthocyanidins, with the major groups of phenolic

**Table 2** Values of some bioactive phytochemicals of lentils

| Bioactive phytochemical                 | Content (per 100 g)        | Reference |
|-----------------------------------------|----------------------------|-----------|
| Polyphenolics, total                    | 26 mg (GAE/100 g fresh wt) | [30]      |
| Flavonoids                              | 221 mg catechin equiv.     | [35, 38]  |
| Glycosides of flavonones                | 33.1–186.0 $\mu$ g         | [33, 35]  |
| Glycosides of flavonoles                | 9.6–241 $\mu$ g            | [33, 35]  |
| Dimers procyanidins                     | 619–1,122 $\mu$ g          | [33, 35]  |
| Trimer procyanidins                     | 441–498 $\mu$ g            | [33, 35]  |
| Procyanidins tetramers                  | 18.5–59.5 $\mu$ g          | [33, 35]  |
| Galloylated procyanidins                | 69.3–123 $\mu$ g           | [33, 35]  |
| Prodelphinidins                         | 369–725 $\mu$ g            | [33, 35]  |
| Glycosides of <i>trans</i> -resveratrol | 5.5–9.3 $\mu$ g            | [33, 35]  |
| Hydroxybenzoic acids                    | 4.5–28.4 $\mu$ g           | [33, 35]  |
| Hydroxycinnamic acids                   | 11.7–29.5 $\mu$ g          | [33, 35]  |
| Catechin                                | 919–1,633 $\mu$ g          | [33, 35]  |
| Gallic acid                             | 140 $\mu$ g                | [33, 35]  |
| Tannins, condensed                      | 870 mg catechins equiv.    | [28]      |
| Phytic acid                             | 620 mg                     | [41]      |
| Saponins, expressed as soyasapogenol B  | 34 mg                      | [41]      |
| Phytosterols, total                     | 22.9–31.6 mg               | [20]      |
| $\beta$ -Sitosterol                     | 123.4 mg                   |           |
| Campesterol                             | 15.0 mg                    |           |
| Stigmasterol                            | 20.0 mg                    |           |

GAE gallic acid equivalent

compounds that are present in the seed coat, but absent in the cotyledon [33–35].

Interestingly, seed coat represents only a small percentage of the entire lentil seed weight, ranging from 8 to 11 %, but provides significant contribution to the overall benefits of lentils [35, 36]. Although it has limited use, lentil hull represents a potential source of value-added healthy products for the expanding dietary fiber market and for further economic phytochemical extraction [29].

It has been found that the darkness of the coat of colored pulses such as lentils is correlated with their phenolic contents. Xu and coworkers [37] found that dark colored pulses such as lentils possess higher phenolic content and antioxidant activities than the pale colored ones. Green or red whole lentils had a significantly higher phenolic content and antioxidant capacity than that of the pale colored pulses [37, 38].

#### Phytosterols

Pulses are one of the major natural sources of phytosterols, along with cereals. Indeed, phytosterols have been shown to be abundant in lentil seeds. Recently, researchers [30] reported that  $\beta$ -sitosterol representing the predominant phytosterol, ranges 15.0–24.0 mg/100 g in cooked dry pulses.

#### Phytic acid

Phytic acid (myo-inositol-1,2,3,4,5,6, hexabisphosphates, IP<sub>6</sub>) is a ubiquitous plant component that constitutes 1–5 % by weight of most pulses [39], and is considered as the major source of phosphorous in their seeds [40]. Nowadays, phytic acid has been proposed to serve a vital role in protecting the seeds against the deleterious effects of oxygen and iron [39]. Lentils are considered good sources for phytic acid and its related phytates [41].

#### Saponins

Saponins include a diverse group of compounds characterized by their structure containing a steroid or triterpenoid aglycone and one or more -ose chains. Lentils, like other pulses, are considered among the best sources of saponins. The lentil content of saponins could be as low as 25 mg/100 g depending on germination conditions [42].

#### Lectins

Lectins or haemagglutinins are a very important group of biologically active proteins or glycoproteins found in almost all organisms [43]. Lectins are found naturally in lentils, with two lectin-binding fractions were isolated by affinity chromatography on immobilized lentil lectin [44].

Chemical structures for the previous functional bioactive components of lentils are found in Fig. 2.

#### Defensins

Defensins form a class of host defense peptides in plants and animals, and participate in the development of the innate immune response. Plant defensins are small basic, cysteine-rich peptides. Recently, defensin has been characterized in germinated lentil seeds [45], and was termed Lc-def. The Lc-def consists of 47 amino acid residues and has 8 cysteines forming 4 disulfide bonds.

#### Protease inhibitors (PIs)

Protease inhibitors have been widely investigated in pulses. Trypsin inhibitors have been identified in lentil seeds with a range of 3–8 trypsin inhibitor unit (TIU)/mg for four different cultivars [46]. Historically, PIs were considered as anti-nutritional components of pulse seeds, due to their property of decreasing the digestibility of dietary proteins.

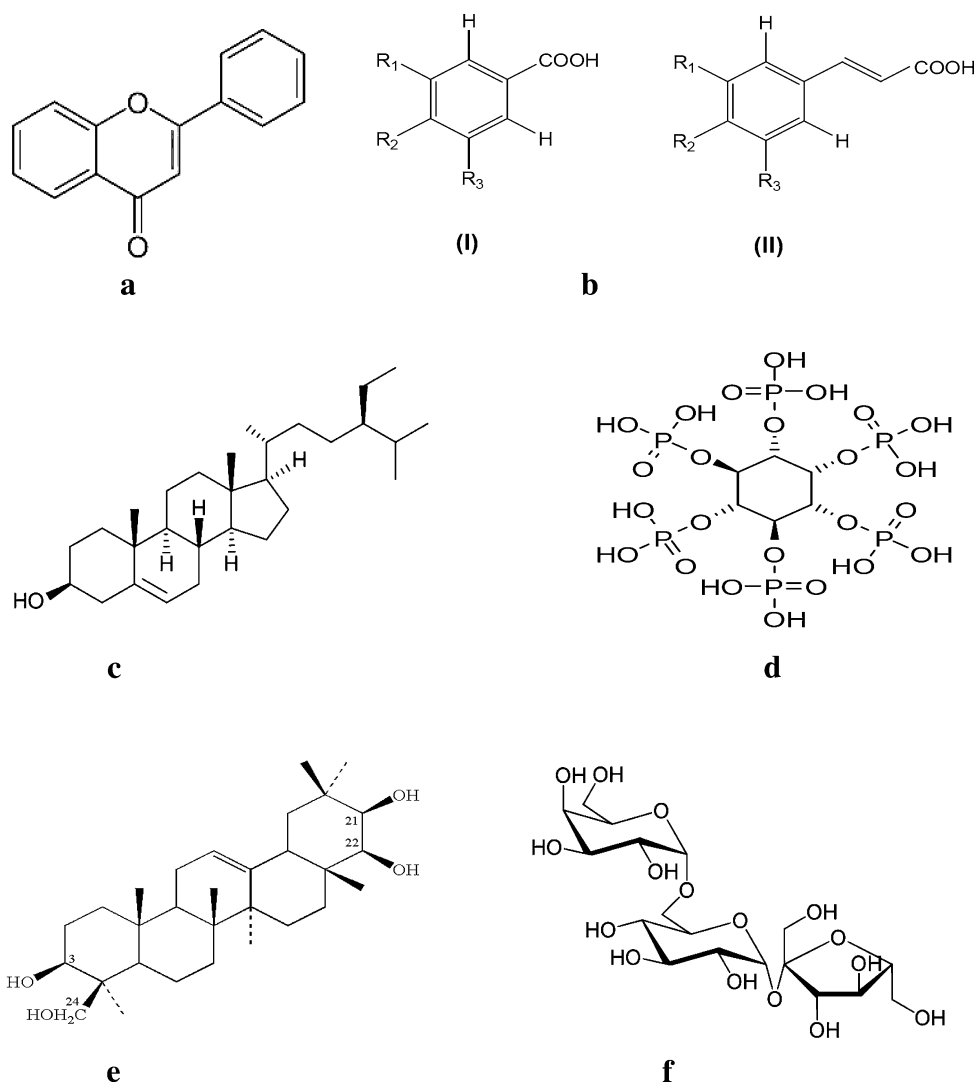
In pulse seeds, Bowman–Birk type trypsin–chymotrypsin inhibitor (BBI) is present at higher concentrations compared to other plant families and tissues [47]. Unlike other pulses, however, kinetic studies have shown that the isolated BBI from *L. culinaris* seeds, *L. culinaris* trypsin inhibitor (LCTI) is characterized by unusual strong binding affinity to its target [48]. In addition; LCTI has been shown to be resistant to thermal denaturation over a wide range of temperature and pH values, which could be due to a generally conserved tertiary structure and hydrogen bond network. Such a unique stability explains the potential beneficial effects of BBIs even after the lentils are cooked [49].

#### Dietary fibers and resistant starches

Lentils could be considered as a valuable source of dietary fibers (Table 1) most of which (93–99.7 %) is insoluble and less than 7 % soluble [12, 17, 50]. The  $\beta$ -glucan component of soluble fibers in lentils is relatively low as compared with its good sources such as oats. However, it is relatively higher than that of peas, winter wheat and flax-seeds [21]. Further, by virtue of their high fiber content, green lentils supplemented to healthy subjects consuming typical western diet had been found to increase their fecal weights significantly, thus aiding in protecting their gut from constipation and its harmful complications [51].

Resistant starches (RS) are those starches that resist the hydrolytic effect of digestive enzymes and could be defined as “the sum of starch and starch-degradation products that, on average, reach the human large intestine” [52]. Cooked

**Fig. 2** Chemical structures for selected functional bioactive components of lentils: **a** flavone backbone of condensed tannins; **b**, (I) hydroxybenzoic and, (II) hydroxycinnamic acid structures; **c**,  $\beta$ -sitosterol structure; **d**, phytic acid; **e**, saponins structure; **f**, raffinose oligosaccharides



lentils contain about 25 g RS/100 g total starch FM, representing about 48 % of total starch content [53], with a value that reaches up to 65.2 % [13]. Due to the presence of RS and other sorts of fibers, lentils and other fiber-containing food exhibited bifidogenic effect even it was low when compared with that of other pulses such as peas and chickpeas [54]. Further, it was found that the indigestible fraction (IF) of lentils is fermented by colonic bacteria, as shown by in vitro fermentation studies [55]. IF from black bean and lentil was the best substrate for the fermentative production of short chain fatty acids (SCFA), especially butyric acid, which are important for the intestinal health [55].

### Antioxidant potential of lentils

Fascinatingly, lentils showed the highest total antioxidant capacity (TAC) among tested pulses measured by

ferric reducing antioxidant power (FRAP) and total radical-trapping antioxidant parameter (TRAP) measures, but came second to broad beans by Trolox equivalent antioxidant capacity (TEAC) measure [56]. These findings were also confirmed by Duenas et al. [36] who found that lentils had a greater antioxidant activity than peas.

In another study, Xu and Chang [57] found that lentils had the highest antioxidant capacity when measured as 2,2-diphenyl-1-picrylhydrazyl (DPPH) free radical scavenging capacity in comparison with green pea, yellow pea and chickpea. The same study also revealed that the oxygen radical absorbing capacity (ORAC) of lentils was significantly higher than that of green pea, yellow pea and chickpea. According to the USDA ORAC values 2007 [58], lentils had a higher ORAC value than most of the common fruits and vegetables including apples, plums, blackberries, cherries, figs, peaches, pears, oranges, garlic, cabbage and almonds.



**Table 3** Health improving effects of lentil components

| Health effect                  | Responsible component (s)                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------|
| Antioxidant                    | Vitamin E, $\beta$ -carotene, polyphenolics                                                    |
| Anticancer                     | Flavonoids, BBI, phytic acid, phytosterols, defensin, lectins, RS, saponins, $\beta$ -carotene |
| Antibiotic                     | Defensin                                                                                       |
| Anti-inflammatory              | Phytosterols, BBI                                                                              |
| Hypolipidemic                  | Phytosterols                                                                                   |
| Reduction of glycemic load     | Resistant starches (RS) when replacing digestible starch                                       |
| Blood pressure-lowering effect | K, proteins                                                                                    |
| Laxative                       | Insoluble dietary fibers, RS                                                                   |
| Bifidogenic                    | Raffinose family oligosaccharides, RS                                                          |

### Health promoting effects of lentils

Interestingly, lentils had been mentioned in ancient treatment remedies and were documented by Dioscorides as one of the therapeutic plants [59]. Lentil seeds are used nowadays in the folk medicine of many ethnicities to treat different illnesses. They are used orally to treat diabetes [60], topically as a water paste to treat skin infections [61] and for the treatment of burns, after being roasted, milled and applied directly to affected areas [62]. In addition, lentils are used as a source of lectins for the treatment and prophylaxis of retroviral infections including human immunodeficiency virus (HIV) infections [63]. Table 3 reviews the health-improving effects of lentils and specify the bioactive constituents responsible for each effect.

#### Lentils and cardiovascular health

Storage proteins of lentils were found by Kingman et al. [64] to exert a hypocholesterolemic effect in pigs. They reported that the underlying mechanism for the hypocholesterolemic effect of pulse feeding does not involve increased cholesterol clearance via the intestinal route, but rather due to the effects on plasma amino acids.

In humans, pulse consumption had been inversely associated with the incidence of cardiovascular diseases (CVD), including hypertension [65, 66]. Recently, Boye et al. [67] found that lentils possess angiotensin I-converting enzyme (ACE) inhibitor activity. The authors concluded that red lentil protein hydrolysates could contribute to the blood pressure-lowering effects of lentils.

Accumulating evidence supports the cardioprotective, hypolipidemic, and hypohomocysteinemic effects of pulses, including lentils [66]. Hyperhomocysteinemia has been linked to an increased risk of CVD. The Framingham Heart study [68] found that lentils contributed 1.7 % of total folate intake. Also, they showed that serum folate

concentrations were significantly increased, whereas serum homocysteine concentrations were significantly decreased in a dose-dependent manner regardless of the source of folate [68]. This finding led to speculation that lentils could exert a beneficial effect in reducing the extent of hyperhomocysteinemia by virtue of their high content of folate, and thus in ameliorating coronary heart diseases (CHD).

In a recent study conducted on 22 adult males with hypercholesterolemia, it was found that daily consumption of folate-rich foods, including 50 g pulses that include lentils and other pulses at a combined amount of 500  $\mu$ g daily for 5 weeks, significantly decreased plasma total homocysteine (tHyc) by 8 % in comparison with consumption of folic acid capsules [69].

In a meta-analysis of eleven clinical trials that examined the effects of pulses on serum lipoproteins, results revealed that intake of non-soya pulses, including lentils, was associated with reduction in fasting serum total cholesterol (TC), lipoprotein cholesterol (LDL), triglycerides (TG) and an improvement in high density lipoproteins (HDL) cholesterol. The reviewers ascribed the hypocholesterolemic effects of pulses to be related, in estimated order of importance, to soluble dietary fiber, vegetable protein, oligosaccharides, isoflavones, phospholipids and fatty acids, saponins and other factors [66].

In our laboratories, lentils were examined in vitro for their glycemic and lipidemic effects using the streptozotocin-induced diabetes rat model [70]. In this study, the administration of lentils significantly increased HDL-cholesterol in diabetic rats. However, the changes have no significant influence on other lipid profile including TG, TC and LDL-cholesterol. This discrepancy in the lipidemic effect of lentils may be due to increase in adipose tissue lipolysis in absence of insulin, and to a decrease in lipoprotein lipase (LPL) activity [71]. To the best of our knowledge and according to the available literature, the underlying mechanism of the hypocholesterolemic effect of lentils had not been studied. However, it has been shown that the hypocholesterolemic effect of another pulse, raw pea, was probably due to increased fecal bile acid output and an increased biliary bile acid formation [72], a matter that could be present in the case of lentils but needs to be investigated.

Among the most compelling effects of pulse carbohydrates are the hypocholesterolemic and CVD risk-lowering effects. The hypocholesterolemic and normolipidemic effects of lentils were studied by several researchers [73–77]. Shams and colleagues [73] found that addition of 50 g cooked lentils to diabetic patient's diet led to a significant decrease in TC but not in LDL, HDL and TG. They explained their findings under the light of the low glycemic index (GI) value of lentils. In such low GI foods, the reduction in blood lipids is thought to be due to the greater

amounts of amylose in comparison with amylopectin. Digestion and absorption of amylose part of starchy foods are much slower than that of amylopectin. Consequently, GI of amylose is less than that of amylopectin, a matter that could explain the blood lipid-lowering effect of lentils which have higher amylose to amylopectin fraction. The hypolipidemic effect of lentils had been ascribed also to increased biliary cholesterol and decreased biliary phospholipids in such a way that the biliary cholesterol saturation index rose from 110 to 169 % [77].

Other researchers [74, 76] have used mixed pulse diets in their studies. Jenkins et al. [74] showed that daily consumption of 140 g of dried lentils and other pulses cooked or canned for 4 months by a group of seven-free living hyperlipidemic males significantly reduced serum TG by 7 % and TC by 25 %, but did not significantly affect LDL- or HDL-cholesterol.

Later, other researchers [75] investigated the effects of lentils and other pulse species on blood lipids and fecal steroids in Sprague–Dawley rats for 8 weeks. They found that lentils were second to Bambara groundnuts in their hypocholesterolemic effect. In addition, their study revealed that plasma TG concentrations were significantly reduced on lentils and other pulse diets, compared to a control western-type human diet, and also plasma LDL-cholesterol concentration was significantly reduced on lentils and other pulse diets, except Bambara groundnuts, compared to control [75].

The latter findings were also confirmed by a controlled interventional study conducted by Duane et al. [76]. In their cross-over study on nine male subjects housed in a metabolic ward for 6–7 weeks, they found that consumption of 120 g baked mixed pulses, 60 % red, navy, and lima beans, 27 % peas and 13 % lentils, significantly reduced serum LDL-cholesterol by 8 % at end of treatment compared to the isoenergetic control diet. However, no other significant change was detected in serum TC, HDL-cholesterol, or very low-density lipoprotein (VLDL)-cholesterol.

It is worth mentioning that because lentils were used as part of a mixed pulse intervention in the latter studies [74–77], the effects of lentils cannot be determined from these studies and the hypolipidemic effect cannot be ascribed solely to lentils. Thus, further research is required to elaborate the impact and mechanism of action of lentils in affecting blood lipids in human subjects.

Bazzano et al. [78] conducted a meta-analysis of randomized controlled trials on the blood cholesterol-lowering effect of non-soy pulse consumption. Ten randomized, selected clinical trials including 268 participants who were given diets rich in non-soy pulses: beans, peas, lentils or chickpeas, for a minimum duration of 3 weeks, showed a significant reduction in total and LDL-cholesterol levels.

They recommended that dietary modification strategies that target the reduction of risk factors for CVD should include an increase in pulse consumption in addition to other strategies, which have been of proven benefit.

### Lentils and diabetes

It has been strongly suggested that eating pulses is beneficial in the prevention and management of diabetes [79]. Therefore, consumption of a wide range of carbohydrate foods from pulses and other rich sources both for the general population and for people with diabetes, especially those with type II diabetes [80] is generally recommended.

Lentil-derived leguminous fibers have been found to prevent the impairment in the metabolic control for diabetic rats when total carbohydrates intake was increased, suggesting that lentil carbohydrates, including their dietary fibers, could have promising implications for diabetic patients [81].

The ability of lentils to alleviate the glycemic load (GL) has been demonstrated in experimentally induced diabetic rats, healthy volunteers, and insulin-dependent and non-dependent diabetic patients [70, 81, 82].

Shams et al. [73] found that addition of 50 g cooked lentils to diabetic patient diet led to a significant decrease in fasting blood glucose. In this study, the administration of lentils significantly decreased serum blood glucose. The glucose-lowering effect of lentils was ascribed by the researchers to probable influences of low GI diets on glucose metabolism, including reduction of glucose poisoning or influence of great amounts of glucose on destruction of pancreas  $\beta$  cells, reduction of proteins and key enzymes glycosylation which are responsible for metabolic processes.

In another relevant study, a group of eight healthy volunteers took a series of breakfast test meals containing either lentils, which had been processed in four different ways, or the same amount of carbohydrate as white bread. Of the different types of lentils, boiled lentils resulted in a flattened blood glucose response in comparison with bread [83].

Glycemic index is a measure of the increase in blood glucose, which occurs after eating a preset portion of available carbohydrate from a test food, compared to a reference food. Consumption of foods with high GI and high GL is associated with hyperglycemia and hyperinsulinemia. Lentils showed low GI value in healthy volunteers, with an average value of approximately 29 [84]. This is within the range reported in the tables of Foster-Powell et al. [85] in which the values ranged from 18 to 52 depending on the type of lentils as follows: Red lentils an average value of 26 (mean of 4 studies) green lentils, an average value of 30 for green lentils (mean of 3 studies), 29 for NS type lentils (mean of 2 studies) and 52 (1 study) for green lentils canned in brine.



There was a noticeable variation in the reported values of GI for lentils between in vitro and in vivo studies [85–89]. Despite this variation between in vivo and in vitro GI values, lentils are still had the lowest and the slowest rate of hydrolysis of their starches, and the lowest expected GI (*eGI*) among tested pulse grains namely chickpeas and peas [89].

#### Lentils and food intake and body weight

The high fiber content and low glycemic response of lentils have been looked at as a means for increasing satiety, reducing the food intake, and thus controlling body weight [90]. Among four different pulses, lentils exhibited the strongest satiating properties, resulting in lower food intake compared to other dietary meals. Lentils led to 8 % (720.1 kJ) lower cumulative energy intake compared to a reference meal [90]. This evidence improves the observational studies that consistently show an inverse relationship between pulse consumption and BMI or risk for obesity [91].

#### Lentils and cancer

Lentils are dietary component traditionally consumed in populations where cancers of the colon, breast, and prostate are low [92]. In a prospective study on 90,630 women, lentils or beans were the only two foods that exhibited an inverse association with the risk of breast cancer [93]. In Turkey, lentils have been considered among the chemopreventive foods for which epidemiological studies revealed to be associated with reduced risk of different cancers [94]. In a case–control design on 186 African–American men and women, results revealed that consumption of pulses, including lentils, was negatively associated with risk of developing colorectal cancer [95]. The reasons behind the association between lentil consumption and reduced cancer risks and incidence could be ascribed to different inherited characteristics and profound constituents for lentils.

Consumption of foods with high GI and high GL is associated with hyperglycemia and hyperinsulinemia. These foods have also been suggested to cause metabolic disturbances correlated with increased incidence of colorectal cancer (CRC) [96]. Hence, the lower incidence of CRC in developing countries compared to developed ones could be partially attributed to the consumption of larger quantities of low GI foods, such as pulses. Lentils have shown significantly the highest polyphenolic content expressed in terms of TPC among different pulses. Polyphenolics have shown chemopreventive ability against cancer with several plausible molecular, genetic and biochemical mechanisms [97].

Plant lectins are unique group of proteins and glycoproteins in lentils with potent biological activity. Several lectins have been found to possess anticancer properties

in vitro, in vivo, and in human case studies [43], with different modes of actions including molecular, genetic and epigenetic mechanisms. The mechanisms by which lectins exert their tumor suppressor effects were summarized by Scarafoni et al. [48]. Previous reports have shown in vitro inhibitory effects of lectins from different plant sources, including lentils, on colon cancer. Roy et al. [4] suggested that most studies investigating anticancer effects of lectins used lectins from lentil and various pea varieties, further confirming the uniqueness of lectins from these pulses compared to other natural sources. Hence, lectins from lentils and other pulses may have great potential to be as functional foods for reducing the risk of certain cancers.

A novel peptide called “defensin” has been characterized recently in germinated lentil seeds [45]. Plant defensins are characterized by a broad spectrum of biological activities including antimicrobial activities against bacteria and fungi. The purified lentil defensin possesses an antifungal activity inhibiting the *Aspergillus niger* growth. Further, several reports demonstrated the ability of defensins to modify the activities of digestive enzymes, trypsin and  $\alpha$ -amylases, and HIV-1 reverse transcriptase enzyme implicated in viral replication. Some defensins have also been shown to block ion channels and inhibit protein translation. Thus, they have been suggested to aid in halting tumorigenesis. Indeed, defensin exhibited an anti-proliferative activity against more than one tumor cell line [45].

Protease inhibitors have been widely investigated in pulses. Among the suggested therapeutic applications of IPs include inhibition of cancer and utilization in acquired immunodeficiency syndrome (AIDS) due to their proteolytic effect against reverse transcriptase enzyme involved in viral replication [48].

Among the bioactive peptides that characterize pulses are the BBI, for which many beneficial biochemical and functional properties have been proved, such as the efficacy of BBI against tumor cells in vitro [48, 98]. Proteases are considered key factors in cancer progression and metastasis; therefore, suppressing their activities by PIs appears to be contributing to inhibiting carcinogenesis [99]. Indeed, BBIs have been shown to possess cancer preventive and suppressing agents in various in vitro and in vivo model systems. A compelling body of therapeutic evidence suggests potential clinical applications for BBI in radioprotection, skeletal muscle atrophy, obesity, autoimmune diseases, multiple sclerosis, and inflammation [99, 100]. Mature lentil BBI were reported to inhibit cell proliferation of colon cancer cells in a dose-dependent manner due to their intrinsic abilities to inhibit serine proteases [100].

Lentils contain considerably high amount of the pivotal folic acid, which is expected to be involved in the cancer preventive effect of lentils. Folate is essential for the de novo biosynthesis of purines and thymidylate, which

affects DNA replication and cell division, and for the synthesis of S-adenosyl methionine (SAM) which is a methyl donor for more than 100 biochemical reactions, including methylation of DNA [101]. Folic acid and SAM work as methylating agents that have the potential to prevent hypomethylation of DNA, which is observed in colorectal neoplasia in humans and may contribute to the loss of normal controls on proto-oncogene expression [102]. Hence, folic acid is considered to be among the most potent agents that suppress gene expression by DNA methylation [103]. In animals, it was found that folate supplementation before the initiation of neoplastic foci significantly decreased the number of small-intestinal adenomas and colonic aberrant crypt foci (ACF) [104]. In humans, higher folate intake has been inversely associated with the risk of colon cancer [103].

Dietary phytates have been shown to be effective in halting colorectal carcinogenesis [105]. The other anticarcinogenic effects of phytates were critically reviewed elsewhere [106, 107]. Furthermore, preliminary studies in human cancer patients showed that IP<sub>6</sub> and inositol, as Ino-Cell<sup>®</sup>, an adjuvant to chemotherapy, appeared to enhance the anticancer effect of the conventional chemotherapy, control cancer metastases, and improve quality of life by reducing the side effects of common chemotherapy [107].

In recent years, food and non-food sources of saponins have been focused upon due to increasing evidence of their positive health implications such as hypocholesterolemic and anticancer properties [42]. Anticancer activity of saponins has been reported for many triterpenes and steroid saponins including, but not limited to, soya saponins present in pulses, including lentils. Different forms of saponins isolated from different plant sources have been identified as potential anticancer agents by the National Cancer Institute's Anticancer Drug Screen Program [42]. Soya saponins have been suggested to be potent chemopreventive agents against CRC, an effect that had been evidenced through several epidemiological, and in vivo and in vitro laboratory studies that were critically reviewed by Gurfinkel and Rao [108].

The presence of a wide spectrum of bioactive phytochemicals and peptides in lentils makes it a functional food with chemopreventive effect against CRC. Thereafter, the chemopreventive potential of lentils against colorectal carcinogenesis was investigated in our laboratory and results were published through two separate articles [109, 110]. In the study, azoxymethane (AOM) was used to chemically induce colon cancer in Fischer 344 rats that were fed different lentil diets before and after the carcinogen injection. Results of the study revealed that lentils significantly reduced the number of dysplastic lesions and neoplasms in the colon of rats when compared with control [109]. Further, ACF, as surrogate endpoints for

chemoprevention trials, the pre-cancerous lesions count, and crypt size and multiplicity were significantly reduced in rat groups fed different lentils in comparison with the control group.

For lentil diets, cooked whole lentils had shown a striking 77.8 % reduction in large ACF, whereas the reduction was only 26.8 % in rats fed raw whole lentils. As large ACF have been shown to predict more accurately preneoplastic potential, this suggested that the lentils, particularly cooked whole lentils, may act by retarding progression of the early aberrant crypts (AC). This preferred effect is elucidated by the substantial reduction in the multicrypt foci, >4AC, by different treatments, and the ability of these dietary factors to considerably reduce the number of aberrant crypts derived from these large foci. Cooked whole lentils resulted in the highest reduction for the AC from large or multicrypt foci, with a percent reduction of about 65 %.

It is noteworthy to indicate that the ability to reduce total ACF number reflects the ability of tested materials to prevent carcinogenesis initiation or working as blocking agents. According to Chen and Kong [103], the blocking agents that inhibit colon carcinogenesis could exert their preventive effect by several mechanisms, including enhancement of detoxification of carcinogens, inhibition of cytochrome P450 (CYP450)-mediated activation of carcinogens, scavenging free radicals and halting antioxidant activity, and finally trapping the carcinogen and preventing their interaction with DNA.

In our study, the chemopreventive ability of lentils in F344 rats was accompanied with increased activity of the xenobiotic detoxifying liver enzymes glutathione-S-transferase (GSTs) in all lentil-fed groups in comparison with the control, with a significant increase in the phase II GST activities in rats fed lentil diets was seen compared to the control. It has been reported that [111] anticarcinogenic enzyme inducers can be either monofunctional which elevates only the phase II enzymes involved in xenobiotic detoxification, e.g., GST, or bifunctional, which elevates both the phase I enzymes that catalyze the metabolic activation of carcinogens, e.g., CYP450, and phase II xenobiotic detoxifying enzymes [111].

Antioxidant polyphenolics present in lentils and other pulses are among the monofunctional inducers that increase the activity of xenobiotic detoxifying enzymes alone [112]. The finding that hepatic levels of antioxidant enzymes such as GST were significantly greater in rats fed chemopreventive lentil diets than in rats fed control diet implies an increased antioxidant capacity to defend against oxidative stress that was triggered by chemopreventive agents [112].

The induced GST in livers of rats fed split lentils could be ascribed partly to the presence of appreciable quantities

of phytic acid in the cotyledons. Phytic acid had been reported to be implicated in elevating hepatic GST levels [113]. In addition, it was found that levels of GST were increased concomitantly with increasing phytic acid, inositol, and polyphenolics content in the tested diets, when compared with the control diet. Phytic acid could reduce colon cancer via chelation of iron and suppression of iron related initiation and promotion of carcinogenesis. Further, it may have potential therapeutic use in cancer due to its property of enhancing the activity of natural killer cells associated with suppressed tumor incidence [113].

Furthermore, the inhibitory action of lentils could be explained, in part, by its putative antioxidant activity, and by the presence of other chemopreventive agents in the cotyledons other than the antioxidant phytochemicals accumulated in the seed coat, as presented in the rat fed the split lentils. In addition, the high selenium content in lentils may also be a contributing factor to anticarcinogenic potential of lentils, as this element is involved in the induction of apoptosis and enhancement of immune system [114]. Thus, it might be speculated that the antioxidant phytochemicals were not the only responsible factors for chemoprevention, as the chemopreventive ability was not restricted to polyphenolic-rich whole, non split, and lentil seeds. Other macro and micro-constituents in lentil cotyledons may explain the reduction in carcinogenesis in rats fed split lentils. However, in our type of research when foods, as opposed to chemicals, are used, it is not possible to identify a specific component of lentils responsible for the observed reduction in early and late colon carcinogenesis.

## Conclusion

When comparing the vast amount of literature investigating lentils, *L. culinaris*, a compelling body of evidence confirms that lentil is one of the most nutritious and health-improving foods known to man. According to recent definitions, lentil could be considered a prophylactic and therapeutic functional food due to its considerable content of essential macronutrients, namely functional proteins and carbohydrates, and essential micronutrients, as well as bioactive phytochemicals such as phytates and polyphenols. Indeed, lentils contain an impressive arsenal of secondary metabolites, minerals and bioactive constituents that have shown to be promising in the management and prevention of several human chronic illnesses. Further research is required to improve the nutritional quality of lentil proteins and optimize the agricultural and culinary conditions to insure maximum utilization of the rich supply of bioactive phytochemicals in lentils. This paper provides a background for researchers to go in depth in exploring the

different molecular and genetic mechanisms involved in the different disease-ameliorating potentials of lentils, namely the anticarcinogenic, hypoglycemic, hypocholesterolemic and blood pressure lowering effects.

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